

Killian Embler

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EDUCATION

- **Embry-Riddle Aeronautical University** Daytona Beach, Florida
Bachelor of Science - Engineering Physics;
Area of Concentration: Spacecraft Instrumentation;
Minors: Applied Mathematics, Computer & Electrical Engineering, Computer Science, Chinese Studies;
CumGPA: 4.00/4.00; Expected Graduation: Spring 2028
Fall 2024 - Present

SKILLS

- **Computer Aided Design:** Inventor, Fusion, Catia, Siemens NX, FEA, EasyEDA, KiCad, URDF Modeling
- **Manufacturing:** Drafting, Tolerances, CNC Lathes, Mills, Welding, 3D Printing (FDM), Laser Cutting, Hand and Power Tools
- **Electrical & Embedded:** Wiring, Soldering, PCB Design, Arduino, STM32, Raspberry Pi, Teensy, FPGA, Oscilloscope
- **Programming:** C/C++, Java, ROS, ARM Assembly, Python, MATLAB, Verilog
- **Organizational:** Word, LaTeX, Excel, Agile Scrum, Experimental Design, Experimentation with Human Participants

WORK EXPERIENCE

- **Robotics Intelligent Learning Lab (Mechanical Engineering Department)** Daytona Beach, FL
Employed Undergraduate Researcher (Part-Time) / Project Lead
Fall 2025 - Present
 - Developing robot-human interactive control interfaces, with focus on assistive exoskeleton systems.
 - Project Lead for both the hip isolated and full lower limb exoskeletons; lead presenter at IROS 2026, submit budget requests, and maintain weekly documentation.
- **Academic Advancement Center** Daytona Beach, FL
Physics III TA / Recitation Tutor (Part-Time)
Fall 2025 - Spring 2026
 - Working alongside the Physics III professor to manage recitation periods, assist in student enrichment.
- **Embler Maritime** West Palm Beach, FL
Marine CAD and Manufacturing Consultant
Fall 2022 - Present
 - Designing and manufacturing wood, acrylic, and 3D printed mechanical components for yachts.
 - Considered component life-cycle feedback to design parts for longevity in high-wear environments with heavy salt and sun exposure.

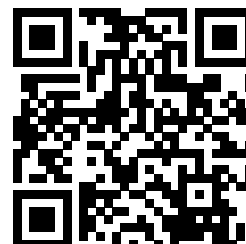
RESEARCH EXPERIENCE

- **Engineering Physics Propulsion Lab (EPPL)** Daytona Beach, FL
Undergraduate Researcher
Fall 2024 - Present
 - Developing externally funded research projects with focus on the OpenMutt quadruped and OpenArm robotic arm.
 - Design, manufacture, and collect test data on robotic platforms, present to and communicate with project supporters to secure funding, develop systems under backer design constraints and time requirements.

PROJECTS

Detailed descriptions, images, and CAD models: killianembler.github.io or scan the QR code below

- **LECTER - Full Lower Limb Exoskeleton: Project Lead.** 14-DOF lower limb exoskeleton for human-robot control research. One of the primary projects of the Robotics Intelligent Learning Lab, contributed to the lab's receiving of a \$523,000 grant from the NIH.
 - Created URDF model, designed mechanical components including dual-linkage ankles for multi-axis actuation, redesigned wiring and power system to battery power with custom PCB and dual E-stop. Tuned impedance controller in ROS2 to assist users in squatting and sit-to-stand motions.
 - First author of an extended abstract accepted to IROS 2026: When Muscles Think workshop, lead presenter of work at the conference.
- **SAMSARA - Flight Computer:** Self-developed rocketry flight computer for Embler Control Systems.
 - 4-layer PCB built around a 32-bit ARM Cortex-M7 processor with an onboard IMU and altimeter, custom integrated DCDC buck converter, and multiple continuity sensing pyro lines, servo ports, and SPI sensor ports.
 - Leading development of firmware for the system.
- **3D Printed Hip Isolated Exoskeleton: Project Lead.** 4-DOF hip exoskeleton as machine learning testbed for metabolic cost reduction research at the Robotics Intelligent Learning Lab.
 - Redesigned mechanical framework with FEA validation, added hip-width adjustability, produced URDF model, and designed the entirety of the electrical architecture including a dedicated PCB.



- **ALTAI - TVC Rocket:** Thrust-vector-controlled rocket developed as an advanced flight computer testbed for Embler Control Systems.
 - Designed and manufactured internal rocket architecture to secure sensitive equipment during flight, deploy a recovery stage electronically without need for pyro lines or deployment charges, and stabilize the rocket without fins via a two-axis gimbal.
 - Led development of stabilization control software using quaternion analysis that saw success across multiple low-altitude launches.
- **Actuator Torque Testbed:** Custom testbed for analyzing torque transmission and mechanical efficiency of actuators under various loads up to 100 Nm. Developed at the Robotics Intelligent Learning Lab.
 - Designed experimental procedure with which to evaluate custom actuators, designed and manufactured adjustable testing apparatus to accommodate multiple different actuators, and configured microcontroller with voltage dividers and pullup resistors to receive sensor signals.
 - Performed experimental procedure for two unique actuators and data analysis via Arduino IDE, Excel, and MATLAB.
- **OpenMutt:** Open-source quadruped manufactured primarily through 3D printing. Developed at EPPL.
 - Redesigned electrical enclosure for compactness, designed precision joint jigs in CAD, fabricated LiDAR mount and auxiliary components, produced URDF model for implementation in ROS2.
- **OpenArm:** Open-source 5-DOF robotic arm designed to mount on the back of OpenMutt. Developed at EPPL.
 - Led mechanical design and URDF creation, and worked on cycloidal gearbox assembly.
 - Redesigned arm at reduced scale upon investor request to be implemented as payload for a Unitree Go2 quadruped. Operated under strict time constraints and investor requirements to perform static analysis of the rescaled arm, spec new actuators, design and manufacture the new arm, and implement a demo in ROS2 for the payload to function at a showcase.
- **Marine Ecosystem Simulation:** Individual software development project as part of object oriented programming coursework.
 - Implemented a genetic algorithm which modeled up to 100,000 unique organisms from more than 50 species and their interactions in a selection of 3 simulated ecosystems.
 - Tracked predation, reproduction, and genetic drift across incremental time periods to determine the impact of primary producer health and presence of invasive species to the long term stability of real marine ecosystems.

LEADERSHIP AND ENTREPRENEURSHIP

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| <ul style="list-style-type: none"> • Embler Control Systems | Daytona Beach, FL |
| <i>Chief Technical Officer & Technical Co-Founder</i> | <i>Sep 2025 - Present</i> |
| <ul style="list-style-type: none"> ◦ Co-founded a 12-person aerospace startup developing advanced flight computers for small to medium scale rocketry. ◦ Coordinated with the PCB design, Software, and Implementation teams on a daily basis during development, advised design decisions, and assisted members with specialized training to increase individuals' confidence and effectiveness. ◦ Spearheaded research and development for the flight computer PCB, hardware implementation of testbeds, organization of launches for experimental validation and product iteration, and firmware for the computer during field operation. (see Projects) ◦ Represented the organization in meetings with elected local officials and representatives of potential business partners, and presented in the Volusia County Pitch Competition, where Embler Control Systems won "Best Technical Startup." ◦ Navigated the process of product development under FCC unintentional radiator constraints, ITAR standards for controlled model rocketry, and FAA safety regulations for test flights. ◦ Accommodated severe global supply shortages in flight computer components such as flash memory, organized the consignment of critical parts from multiple suppliers to a single point of assembly, and minimized production cost through advance component selection considering budget constraints and performance requirements. | |

INVOLVEMENT

- **ERAU:** Engineering Physics Propulsion Lab
- **ERAU:** Robotics Intelligent Learning Lab
- **ERAU:** Society of Physics Students
- **ERAU:** Astrophysics Research Club
- **EXTERNAL:** Sigma Pi Sigma - The National Honors Society for Physics and Astronomy
- **EXTERNAL:** Daytona Regional Chamber of Commerce

AWARDS

- **ERAU:** Diamond Eagle Scholarship
- **ERAU:** Paul B. Hunter Memorial Endowed Scholarship
- **NASA:** NASA Florida Space Grant Recipient - for Exoskeleton research under the Robotics Intelligent Learning Lab
- **Volusia County Pitch Competition:** Best Technical Startup
- **ERAU:** Dean's List - Fall 2024 until Present
- **ERAU:** Spark Grant - for OpenArm robotic arm research
- **National Institutes of Health:** NIH Grant of \$523,000 - awarded to Robotics Intelligent Learning Lab for exoskeleton research, including the LECTER project